NASM VIRTUAL COACHING SPECIALIST (VCS)

# Chapter 7: Integrating Wearables Into Virtual Coaching

National Academy of Sports Medicine — Evidence-Based Telehealth Curriculum

Chapter 7Integrating Wearables Into Virtual Coaching
Lesson 1Introduction to Fitness Wearables

Wearable technology unlocks the ability to incorporate objective biometrics related to daily activity, sleep, recovery, and health parameters to develop and dynamically adjust integrative training programs. This chapter will focus on education and validation of health and performance for common wearable biometrics, the development of data-driven programs, and how to leverage wearable data for compliance monitoring and client reinforcement to maintain motivation and retention.

## Learning Objectives

After completing this chapter, the NASM Virtual Coach will have the ability to do the following:

- **Develop** an understanding of wearable technology and its limitations in virtual training.
- **Summarize** the potential value of integrating wearable devices for virtual training.
- **Describe** the primary biometrics collected by commercial wearable devices.
- **Determine** ways to use biometric data to develop individualized programs.
- **Define** ways to make data-driven adjustments to optimize safety, outcomes, and compliance.

# Introduction

According to Vogels (2020), “Roughly one-in-five U.S. adults (21%) say they regularly wear a smartwatch or wearable fitness tracker” (**Figure: One-In-Five**). When available, wearable technology provides fitness professionals with the ability to improve the reliability of data collection and continuously monitor clients’ health and fitness **biometrics** in a virtual training environment. The data from wearable devices can be utilized for assessing client risk and readiness for physical activity, developing individualized programs, monitoring progress, and even as a reinforcement tool to optimize adherence and retention.



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FigureOne-In-Five
Biometrics

Measurements of the human body or characteristics, including those obtained from smartwatches and wearable devices.

# Defining Wearables

With the use of **wearables**, the virtual coaching experience does not have to be limited to an initial assessment, intermittent communication, and subjective reporting. When integrated properly, wearable technology helps to facilitate a 24/7 monitoring approach. This approach can improve client outcomes while also optimizing a fitness professional's ability to scale his or her virtual coaching business.

While the perception of wearable technology is often limited to wrist-worn devices, this industry includes any type of electronic device worn on the body. Other common wearable devices include smart jewelry, tech-enabled clothing, and eyewear. For the purposes of virtual coaching, this chapter will limit its focus to commercially available smartwatches and fitness trackers that collect and track applicable biometrics including the following:

- Heart rate
- Heart rate variability
- Respiratory rate
- Oxygen saturation
- Physical activity
- Sleep parameters

To collect physiological data, smartwatches and fitness trackers often come equipped with sensors, including an electrocardiogram (ECG) or photoplethysmogram (PPG), gyroscope, and accelerometer.

Description of Wearable Sensors

| Sensor Type | Description |
|---|---|
| Electrocardiogram (ECG) | Direct measurement of the heart's electrical activity |
| Photoplethysmogram (PPG) | Optical measurement of blood circulation obtained by illuminating the skin and measuring changes in light absorption |
| Gyroscope | Measurement of orientation in space |
| Accelerometer | Measurement of three-axis movement and acceleration |

Wearable

Any type of electronic technology worn on the device, including smartwatches, fitness trackers, smart jewelry and tech-enabled clothing.

## Differences in Wearable Devices



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Wearable technologies have many differences, including the type of embedded sensors, method of data collection, what they measure, advanced features, and the quality and reliability of the data. When integrating wearable devices into a virtual coaching business, fitness professionals should be able to adapt and work with a device already owned by a client, but also be prepared to make suggestions based on the most suitable option for their needs.

Getting Technical

**Electrocardiography (ECG) Versus Photoplethysmography (PPG) in Wearable Technology**

Electrocardiography (ECG) is a measurement of electrical activity at the heart that controls the expansion and contraction of the heart chambers, often using multiple electrodes. ECG is the gold standard measurement used in hospitals and healthcare settings to monitor heart rate and heart rate variability.

Photoplethysmography (PPG) uses light-based technology to measure the rate of blood flow as controlled by the heart's pumping action. In addition to measurements of heart rate, analysis of the PPG waveform can provide an accurate estimation of additional biometrics, depending on the wavelength or color of light utilized.

**FIGURE: ECG vs. PPG**



FigureECG vs. PPG

### Green Versus Red Light PPG

Most wearable devices utilize green light PPG. Measurements done on light-skinned individuals at normal temperature have shown that green light has desired properties for data collection, particularly during motion (Maeda et al., 2011). Green light PPG has been shown to be more readily absorbed by individuals with higher melanin levels (Shcherbina et al., 2017), including those with darker skin tones, which can often lead to inaccuracies in data collection.

Red and infrared (IR) light PPG sensors, while less reliable during physical activity, have various benefits related to deeper penetration in the skin (**Figure: Green/Red Light PPG**) and typically provide more reliable data regardless of skin tone (Spigulis, Gailite, Lihachev, & Erts, 2007). To correct for inaccuracies caused by motion in red light PPG sensors, many wearable technology companies have created algorithms that utilize the accelerometer to produce high-quality PPG signals. For these adaptive filters, the accelerometer movement data is used to detect and subtract motion from the PPG signal or to exclude the data completely if an accuracy threshold cannot be obtained.



FigureGreen/Red Light PPG

### Advanced Features

The features and biometric data collected from wearable devices will vary by manufacturer and even by device offered by the same wearable manufacturer. The number of advanced features often increases with the cost of the device, including color screens, onboard GPS, and waterproofing. More expensive is not always better, as many features may be unnecessary or lead to additional confusion for the client. Fitness professionals must understand what each device offers. They must know how the device relates to their ability to develop and execute effective virtual coaching and consider the balance between cost, features, and ease of use for the client.

### Data Reliability



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When relied on for **risk stratification**, program development, and client monitoring, the accuracy of data collected is of the utmost importance. In a comparative study of six mainstream wearable fitness trackers, researchers found that "Fitness trackers of different brands vary with regard to the measurement of indicators and are all affected by the activity state" (Xie et al., 2018).

All wearable technologies will have some margin of error relative to costly and bulky medical equipment. However, using the same device consistently will allow the client to establish a baseline activity level and show the fitness professional a developing trend to support improvements and adjustments. In general, fitness professionals should select a reliable device, avoid switching between different devices, and focus on establishing and monitoring data trends over time.

Coach's Corner

**Selecting a Reliable Wearable Technology**

When determining which wearable technology to integrate into a virtual coaching business, fitness professionals should research which devices have been clinically validated versus gold standard medical equipment. For example, heart rate and/or oxygen saturation should be validated against a hospital-grade electrocardiogram or pulse oximeter and not against other commercial devices lacking validation.

Most commercial wearable technologies do not have the approval of the U.S. Food and Drug Administration (FDA approval) but may have a research section available on their company website. Additionally, fitness professionals may search for scholarly articles on Google Scholar, JSTOR, or a comparable resource for information about data reliability of certain devices.

Risk stratification

Assessment of risk based on qualifying information, such as vitals or health history – Risk stratification can include the potential for training-related injury, the likelihood of a health event, or developing a health condition.

## Pros and Cons

Pros and Cons of Wearable Devices

| Category | Pros | Cons |
|---|---|---|
| Electrocardiography (ECG) | Clinical gold standard | Require multiple electrodes; diminished accuracy when measured at the wrist alone |
| Photoplethysmography (PPG) | Cost-effective; less invasive; provides an estimation of additional biometrics | Less motion-tolerant |
| Green Light PPG | More motion-tolerant | Limitations for certain client populations |
| Red Light PPG | Deeper penetration; reliable for broad client populations | Less motion-tolerant |
| Advanced Features | May increase value for certain client types | Complexity of use; more expensive |
| Data Reliability | Confidence in program development and decision making | N/A |

While wearable technology presents the opportunity to improve a virtual coaching experience, fitness professionals should not rely solely on biometric data. The trainer–client connection should include communication and consideration of subjective feelings and opinions, in addition to objective data obtained from wearable devices.

# Biometric Education

Biometrics are measurements of the human body or characteristics. These measurements range from fingerprint recognition to more readily available measurements obtained from smartwatches and fitness trackers. To understand how to integrate biometric data into a virtual coaching business, fitness professionals should first have an in-depth understanding of that data by considering these questions (**Figure: Evaluating Biometrics**).



FigureEvaluating Biometrics

## Heart Rate



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Heart rate, measured in beats per minute (bpm), is the number of times the heart beats per minute. Heart rate is most commonly assessed at rest and during physical activity, both of which should be considered independently. According to the American Heart Association (2016), the following factors may influence heart rate:

- Fitness Level
- Room Temperature
- Body Position
- Emotional Stress
- Body Size or Composition
- Use of Medications

Critical!

**The Role of a Fitness Professional**

A fitness professional is not credentialed to diagnose, treat, or prevent any specific medical conditions. While many of these biometrics may provide insight into potential health issues, the role of wearable technology in virtual coaching is to provide reliable data for program development. Fitness professionals should not provide medical advice but, rather, recommend that clients seek medical attention from a qualified health professional if alarming biometric values or trends are recognized, especially if accompanied by symptoms including chest pain, shortness of breath, dizziness, or fatigue.

### Resting Heart Rate

**Resting heart rate (RHR)** is the number of beats per minute obtained during rest. A normal resting heart rate can vary widely from person to person, but the accepted range for adults is 60 to 100 bpm (American Heart Association, 2015) (**Figure: Resting Heart Rate**).



FigureResting Heart Rate

However, individuals with higher resting heart rate values within this range may still be at increased risk of cardiovascular, non-cardiovascular, and all-cause mortality (Seccareccia et al., 2001). An elevated resting heart rate above 100 bpm, referred to as tachycardia, can be indicative of an underlying health condition that should be addressed by a trained healthcare professional.

In general, a lower resting heart rate is related to improved heart function and cardiovascular fitness (Nauman et al., 2012). A well-trained individual may maintain a resting heart rate well below 60 bpm, referred to as bradycardia, due to positive cardiovascular adaptations from training. For clients who are not physically active but still maintain a heart rate below 60 bpm, fitness professionals should prompt the client to seek approval from a doctor prior to engaging in physical activity, as this can be a sign of an underlying heart condition (American Heart Association, 2016).

When assessed properly, resting heart rate should be relatively stable for all populations with minimal day-to-day changes. Over time, positive or negative lifestyle interventions may lead to a steady change in resting heart rate values. In a meta-analysis, Reimers et al. (2018) found that all types of exercise, especially endurance training and yoga, decreased resting heart rate in healthy subjects, which may contribute to a reduction in all-cause mortality.

Resting heart rate (RHR)

The number of times the heart beats per minute while at complete rest.

### Active Heart Rate

**Active heart rate** is the number of beats per minute obtained during physical activity. In general, active heart rate is correlated with relative output and can be utilized to gauge the intensity of an exercise effort (**Figure: Target Heart Rate**). Fitness professionals may utilize active heart rate monitoring as a method to assess compliance for exercise prescription, particularly when target heart rate zones are recommended (Polar, 2020).



FigureTarget Heart Rate

Many wearable devices rely on active heart rate measurements, in addition to demographic information, including gender and age, to estimate activity-related caloric expenditure.

Active heart rate

The number of times the heart beats per minute during physical activity.

## Heart Rate Variability

**Heart rate variability (HRV)** is the measurement of variation in time intervals between heartbeats, most commonly reported in milliseconds (ms). Wearable devices are frequently limited to time-domain measurements of HRV and report values as the standard deviation of N-N intervals (SDNN) or root mean squared standard deviation (RMSSD). Additional measures of HRV include nonlinear or frequency-domain analysis, which is beyond the scope of this discussion.

Time-Domain Measurements of HRV

| Index | Units | Description |
|---|---|---|
| SDNN | ms | Total variability between heartbeats, considered the gold standard when measured over a 24-hour period to predict morbidity and mortality (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996) |
| RMSSD | ms | Beat-to-beat variance in heart rate to estimate parasympathetic influences (Shaffer et al., 2014) |

HRV provides quantifiable and non-invasive insight into the current state of a client's **autonomic nervous system**, which is involved in the regulation of autonomic or involuntary processes. The autonomic nervous system consists of two primary branches, sympathetic (fight-or-flight) and parasympathetic (rest, digest, and relax). HRV is a measurement of the ongoing contention between these two branches to regulate physiological processes without conscious thought. For example, an individual does not have to actively think about breathing, their heart beating, or maintaining blood pressure due to autonomic control of these processes. More importantly, though, recent research has shown that changes in HRV may be related to inflammatory processes, hormone regulation, and immune system function (Liu et al., 2017).



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Heart rate variability (HRV)

The measurement of variation in time intervals between heartbeats, most commonly reported in milliseconds (ms).

Autonomic nervous system

The division of the peripheral nervous system responsible for regulation of involuntary processes that do not require conscious control, such as heart rate or respiratory rate.

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Critical!

**Heart Rate Variability and Health**

Due to its relation to physical and mental health outcomes, HRV is among the most meaningful biometrics that fitness professionals should consider when developing and measuring the success of a lifestyle intervention program. There is growing evidence that various diseases are accompanied by reduced HRV values including, but not limited to, the following (Young & Benton, 2018):

- Diabetes
- Cardiovascular disease
- Psychiatric disorders
- All-cause mortality

In a meta-analysis, researchers found that individuals with lower HRV values compared to their cohort had a 112% higher risk of all-cause death and a 46% increased incidence of cardiovascular events (Fang et al., 2019).

Heart rate variability is a highly individualized biometric and should not be utilized to compare different clients but, instead, assessed for positive or negative change at the individual level. While HRV has been shown to decrease predictably with age (Jandackova et al., 2016) and various health-related conditions, the value of monitoring HRV in a virtual coaching program is to determine an individual's recovery, daily training readiness, and monitor progress over time. In general, HRV values should be relatively stable from day to day but may increase gradually over time with positive lifestyle interventions, including improved sleep, exercise activity, and nutritional habits. Higher HRV values have been positively associated with improved cardiovascular fitness and stress resilience (Dong, 2016) and decreased risk of training-related injuries (Williams et al., 2017). Additional studies have shown that decreased heart rate variability is an effective tool for measuring cumulative physical fatigue in overtrained athletes (Pichot et al., 2000), which may offer insight into programming periods of lower intensity to prioritize recovery (**Figure: Heart Rate Variability**).



FigureHeart Rate Variability
Coach's Corner

**Guiding Training With HRV**

HRV represents the current state of the individual's autonomic nervous system and its ability to make a positive adaptation. By assessing an individual's HRV trend and adjusting daily training efforts accordingly, fitness professionals can help clients train smarter while also improving overall outcomes.

**Training Prescription Guided by HRV in Cycling** (Javaloyes et al. 2018)

In a study of elite cyclists, researchers divided participants into two groups: an HRV-guided training group and a traditional periodization group. The HRV-guided training group had their daily output dynamically adjusted based on daily HRV values, whereas the traditional periodization group followed a structured protocol. Following an 8-week training intervention, researchers found that HRV-guided training led to greater increases in the following:

- Peak power (5%)
- Power at second ventilatory threshold (VT2) (14%)
- Power over a 40-minute time trial (7%)

## Oxygen Saturation

When an individual breathes, oxygen is attached to and distributed throughout the body by a protein molecule called **hemoglobin** in the red blood cells. **Oxygen saturation**, often referred to as SpO2 and reported as a percentage (%) value, is the fraction of oxygenated hemoglobin relative to the total amount of hemoglobin. The normal range for oxygen saturation in healthy adults is 95 to 100 percent at rest (**Figure: Normal Oxygen Saturation**) but can be as low as 88 to 92 percent in clients with known conditions such as Chronic Obstructive Pulmonary Disease or other lung diseases (Holland, 2019). Lifestyle habits such as smoking may also reduce oxygen saturation (Ozdal et al., 2017).



FigureNormal Oxygen Saturation

Monitoring oxygen saturation is important because low blood oxygen (hypoxemia) can lead to adverse effects on important organ systems, including the brain, heart, and kidneys (Hafen & Sharma, 2020). In general, clients with normal oxygen saturation should not recognize significant day-to-day changes or overtime with program interventions. If a significant decrease in oxygen saturation is recognized, particularly if it is accompanied by symptoms, fitness professionals should urge clients to seek medical attention.

Hemoglobin

A red protein responsible for transporting oxygen in the blood of vertebrates.

Oxygen saturation

Often referred to as SpO2 and reported as a percentage (%) value, the fraction of oxygenated hemoglobin relative to the total amount of hemoglobin.

## Respiratory Rate

**Respiratory rate**, measured in breaths per minute (bpm), is the number of breaths an individual takes in 1 minute at rest. Respiration is measured as one complete inhale and exhale, which represents the cycle of respiratory gas exchange of oxygen and carbon dioxide. Normal ranges for respiratory rates vary throughout the life cycle (**Figure: Respiratory Rates by Age**), with newborns breathing at a rate of up to 60 breaths per minute but leveling out at a steady rate of 12 to 20 breaths per minute in adulthood (Nall, 2017). For adults, respiratory rates below 12 or over 25 are considered abnormal and may be indicative of an underlying condition such as asthma, anxiety, pneumonia, heart, or lung disease (Cleveland Clinic, 2019). Respiratory rate may also be influenced by posture or mechanical restrictions of the diaphragm (Zafar et al., 2018).



FigureRespiratory Rates by Age

Based on physical activity and other stressors, respiratory rate can vary significantly throughout the day. Fitness professionals should focus on nocturnal measurements of the respiratory rate due to the relatively controlled environment. In general, the nocturnal respiratory rate should be steady from day to day with minimal change over time. Minor changes in nocturnal respiratory rate may be indicative of recent alcohol consumption (Izumi et al., 2005), whereas rapid and significant changes may be indicative of illness requiring medical attention (Cretikos et al., 2008).

Coach's Corner

**Assessing Respiratory Rate During Activity**

In response to physical activity, the autonomic nervous system will increase the respiratory rate to account for increased oxygen demand and a change in blood pH due to the increased concentration of carbon dioxide. Changes in respiratory rate measured during physical activity have been shown to be a valid and reliable marker for assessing cardiorespiratory fitness and anaerobic threshold during a graded exercise test (Carey et al., 2005), which may be valuable for program development. Unfortunately, most wearable devices are intolerant to motion and do not provide a reliable measurement of respiratory rate during physical activity.

Respiratory rate

The number of breaths an individual takes in 1 minute at rest, typically measured in breaths per minute (bpm).

## Sleep Parameters



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Adequate, quality sleep is an important yet overlooked aspect of health and physical performance. According to the Centers for Disease Control (Colton & Altevogt, 2006), one-third of adults in the United States have insufficient sleep. This correlates to wide-ranging effects on the cardiovascular, endocrine, immune, and nervous systems, leading to an increased risk of health conditions, including the following (Center for Disease Control, 2020):

- Obesity
- Type II diabetes
- Hypertension
- Cardiovascular disease
- Anxiety and depression
- Decreased testosterone

General sleep recommendations will vary by age group but may not be appropriate for all client populations depending on health status and level of physical activity. Fitness professionals should monitor the client's sleep duration in relation to other biometric parameters, including sleep quality and heart rate variability to determine more precise sleep recommendations within the appropriate guidelines (**Figure: Sleep Recommendations**).



FigureSleep Recommendations

**Sleep, Recovery, and Adaptation**

While sleep seems like a time to finally let the brain "shut off," research has revealed the brain works even harder during sleep. During sleep, the brain works to restore and repair cells, process information for long-term memory, and perform important processes related to health and performance (Dattilo et al., 2011).

Fitness professionals should emphasize the importance of proper sleep for recovery and dynamically modify programs to optimize client outcomes and decrease the risk of injury.

Wearable devices provide fitness professionals with the ability to consistently monitor clients' sleep habits to determine deficiencies and develop data-driven programs for improvement. Wearable devices auto-capture meaningful sleep parameters and often provide an overall sleep score for rapid interpretation of change over time.



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Common Wearable Device Sleep Parameters

| Sleep Parameter | Description |
|---|---|
| Time in Bed | Total time spent in bed, which may include time spent watching television or on waking before getting prepared for the day |
| Sleep Duration | Total time spent asleep – sleep duration relative to the time in bed can be utilized to assess sleep efficiency |
| Onset and Wake Time | Time of sleep onset and wake up time – these values can be utilized to assess consistency in sleep schedule or adherence to a circadian rhythm |
| Sleep Stages | Analysis of time or percentage of time spent in each distinct sleep stage – sleep stages vary by wearable device but often include light sleep, deep sleep, and REM sleep |
| Latency | The amount of time to fall asleep |
| Awakenings/Disruptions | The number of awakenings, which may include voluntary awakenings such as using the restroom or involuntary, including a sleep disruption |

Additional valuable sleep parameters provided by certain wearable devices include movement, snoring levels, circadian rhythm scores, sleep accounting, and overall sleep scores ranging from 0 to 100 for simple interpretation. Each reported sleep metric provides a fitness professional with the opportunity to develop an individualized and data-driven program to improve sleep quality for overall program success.



FigureOverview of Sleep Stages
Getting Technical

**Introduction to Sleep Stages**

There are four distinct stages of sleep each with the purpose of restoring or improving specific physiological processes. In general, the body will cycle through all of these stages several times throughout a single night, but various lifestyle factors, ages, and health conditions can impact the amount of time spent in each sleep stage.

**FIGURE: Overview of Sleep Stages**

Critical!

**Circadian Rhythm**

One of the most meaningful, simple, and cost-effective (free) interventions that a fitness professional can recommend is adherence to a consistent sleep cycle. The human body has evolved to follow a natural and internal 24-hour clock, or circadian rhythm, which impacts body temperature, metabolism, and hormonal processes (National Institute of General Medical Sciences, 2020). Physiological processes are optimized when factors such as sleep and wake up times are consistently maintained. Consistency in sleep patterns has been correlated with improved cognitive function and memory, whereas inconsistent circadian patterns have shown to increase the incidence of illness, cardiovascular disease, and cancer (Scheiermann et al., 2013). The human body's natural circadian rhythm can be disrupted by external factors, including temperature and light.

Fitness professionals may consider circadian rhythm interventions or recommendations to maintain consistent sleep onset and wake up times to improve an individual's overall sleep quality and performance.

## Recovery and Readiness

Recovery and readiness to engage in physical activity are relative concepts that require access to consistent and reliable biometric data. For example, a fitness professional cannot compare two individuals side-by-side and determine who is more recovered based on a heart rate variability value. Instead, it requires an understanding of an individual's baseline values over the previous days or months to determine the current state of their body. Regarding the common biometrics provided by wearable devices, fitness professionals should assess recovery and readiness based on the biometrics that have been clinically validated to predict the risk of injury, daily readiness to perform, or change in health status. These biometrics are resting heart rate, heart rate variability, and sleep parameters (**Figure: Assessing Recovery and Readiness**).



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FigureAssessing Recovery and Readiness

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Example of Adequate Recovery – READY

| Biometric | 30 Day Average | Today | Interpretation |
|---|---|---|---|
| Resting Heart Rate | 72 bpm | 72 bpm | Normal |
| Heart Rate Variability | 54 ms | 58 ms | Above Average |
| Sleep Duration | 7 hours 48 mins | 8 hours 10 mins | Adequate Sleep |

In the example provided in **Table: Example of Adequate Recovery – READY** the client appears to have had a restful night's sleep and made a full recovery from the previous day. The client may engage in physical activity as planned.

Example of Inadequate Recovery – RECOVER

| Biometric | 30 Day Average | Today | Interpretation |
|---|---|---|---|
| Resting Heart Rate | 72 bpm | 74 bpm | Elevated HR (within normal range) |
| Heart Rate Variability | 54 ms | 40 ms | Below Average |
| Sleep Duration | 7 hours 48 mins | 5 hours 10 mins | Inadequate Sleep |

In the example provided in **Table: Example of Inadequate Recovery – RECOVER**, the client did not achieve adequate sleep and biometrics have not fully recovered, indicating a potential increased risk of injury or poor performance (Williams et al., 2017; Buchheit et al., 2009). The fitness professional should consider modifying the program to focus on active or passive recovery and recommend that the client get a restful night's sleep.

For the purposes and simplicity of these examples, sleep duration was utilized as a primary biometric to assess recovery and readiness. Note that sleep duration alone should not be relied on to determine sleep quality and readiness. Additional parameters, including sleep stages and disruptions, should be included in the determination of sleep quality.

Following a defined baseline stage, many wearable applications apply unique algorithms to provide a recovery score ranging from 0 to 100 for rapid interpretation and simple understanding. These often include basic instructions to rest and recover, take it easy, or engage in light physical activity or indicates readiness to engage in rigorous physical activity based on full recovery.

Recovery Template

# Building a Program

The value of integrating wearable technology into virtual coaching is not from the data collection itself but the insight this data provides to the fitness professional for individualized and data-driven program development. When used properly, wearable technology and the data it provides can be utilized to develop programs and make data-driven adjustments that optimize outcomes and safety. Wearable technology data can also provide a behavioral reinforcement tool for ongoing motivation, adherence, and retention.



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## Process Overview

To optimize the efficiency and value of integrating wearable technology into virtual coaching, the fitness professional should develop a consistent and repeatable process (**Figure: Program Design**).



FigureProgram Design
Integrating Wearables Into Virtual Coaching

| Phase | Overview/Description |
|---|---|
| 1 – Baseline | Establish a minimum of a 3- to 5-day baseline to obtain objective insight into a client's biometric strengths and deficiencies. During the baseline phase, the client should be instructed to continue with normal lifestyle behaviors and not adjust nutritional or exercise habits. |
| 2 – Analysis and Initial Program Development | Following the baseline phase, fitness professionals should perform a comprehensive analysis of a client's strengths and deficiencies in relation to their goals. Programs should be developed to address deficiencies that may limit program success while ensuring that strengths are not sacrificed. A comprehensive program is developed and provided to the client with an explanation of WHY specific interventions were chosen. |
| 3 – Ongoing Monitoring | Fitness professionals should develop a consistent schedule for monitoring client data. While it may not be feasible to assess data every day, it is important to check in on days that rigorous training is scheduled to assess readiness and, if necessary, adjust program recommendations. Some wearable platforms provide automated alerts of significant change to streamline the monitoring process.<br><br>Utilize biometric data to dynamically adjust program recommendations based on client recovery and readiness to safely and effectively perform the recommended program for the day. Over time, fitness professionals may be able to create a more effective program schedule based on recovery trends from previous exercise interventions. |
| 4 – Adjust & Consult | If a program adjustment is required, fitness professionals should attempt to understand the unexpected change in biometric data. For example, poor sleep quality and decreased HRV may be explained if the client consumed alcohol the evening prior. This represents an opportunity to show the client how this behavior impacted their biometrics to create awareness and avoid future deviations from the program.<br><br>In the case that positive biometric trends are recognized, this represents an opportunity to provide positive reinforcement to maintain program adherence. |

## Intervention Toolkit

Fitness professionals should have a comprehensive understanding of the biometrics provided by wearable technologies and develop an intervention toolkit based on their skills and qualifications. This may be organized based on which biometric it is validated to address and then selected on a client-by-client basis based on analysis of additional assessments, cost, and time commitment. Over time, the toolkit will be enhanced and refined based on personal experiences and program outcomes.

Coach's Corner

**Program Development – Practical Example**

Client A is a 45-year-old male with a desk job and a high-level of work-related stress. After a check-up from his doctor, Client A was diagnosed as pre-diabetic and obese. Client A's physician cleared him for exercise but emphasized the importance of lifestyle changes. Client A has tried to exercise and improve his nutritional habits but gives up because the scale weight does not improve.

Following a comprehensive assessment and establishing a 5-day biometric baseline, the fitness professional identified the following deficiencies:

- 2400 steps/day
- Sleep latency over 30 minutes
- Lack of consistency in sleep times
- Elevated resting heart rate (88 bpm)
- Reduced HRV (28 ms)

Based on the fitness professional's qualifications, it is determined that a minimally invasive and phase-based program will be the most effective and sustainable for Client A:

- Daily step count goal of 5000 steps, increased by 500 steps per week: To emphasize consistency of movement throughout the day, Client A is instructed to set an alarm and take at least 500 steps every hour during the workday.
- Exercise program (based on other assessments)
- Nutritional counseling (based on nutritional assessment forms)
- Sleep recommendations and Circadian Rhythm Intervention: Avoid use of technology for 1 hour prior to bed, create a bedtime routine, and set consistent sleep and wake times each day.
- Education: Why these biometrics are important, expected improvements, and the significance to Client A's health and goals

The fitness professional will monitor Client A's biometric data for compliance and consistently provide positive reinforcement based on improvements in sleep quality, daily physical activity, and other applicable biometrics.

## Data-Driven Adjustments

Human physiology is complex and adaptable. This requires fitness professionals to consistently monitor and dynamically adjust programs to minimize the risk of training-related accidents, avoid injuries, and optimize client outcomes. The use of wearable technology provides fitness professionals with the unprecedented ability to objectively quantify and track the efficacy of any type or combination of interventions. For example, a fitness professional with the proper qualifications may develop a similar nutrition program for two individuals who present with similar backgrounds and biometrics but recognize the high probability of significantly different health and body composition outcomes.



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The following are examples of various data-driven adjustments:

- Adjusting the type or intensity of a single exercise session based on recovery data
- Adjusting a portion of an intervention to test if outcomes can be improved, such as modifying macronutrient recommendations
- Removing an intervention that does not seem to be yielding outcomes
- Removing an intervention that is negatively affecting biometric data
- Adding an intervention based on changes in biometric data over time
- Dynamic modification of a program to avoid plateaus

For example, a previously sedentary client following an exercise intervention may require additional sleep to facilitate adequate recovery.

## Behavioral Considerations

The basic principles of human behavior state that behaviors that are reinforced tend to be maintained. Clients have likely attempted lifestyle modification on their own prior to seeking help from a fitness professional but failed because they did not recognize improvements and could not maintain intrinsic motivation. Utilizing biometric data from wearable devices provides fitness professionals with the ability to consistently monitor compliance, which is a notable limitation of virtual coaching, and provides a source of **positive reinforcement** to optimize ongoing momentum for improvement (**Figure: Reinforcement Theory of Motivation**). Fitness professionals should actively pursue any opportunity to make a client feel good about their progress, whether it be based on compliance to a program or positive biometric trends related to overall health and performance (Expert Program Management, 2018).



FigureReinforcement Theory of Motivation

Additionally, biometric data can be utilized to bring attention to the negative impact of certain behaviors. For example, a decrease in sleep quality and heart rate variability following a night of heavy alcohol consumption may decrease the likelihood of the individual deviating from the program in the future.

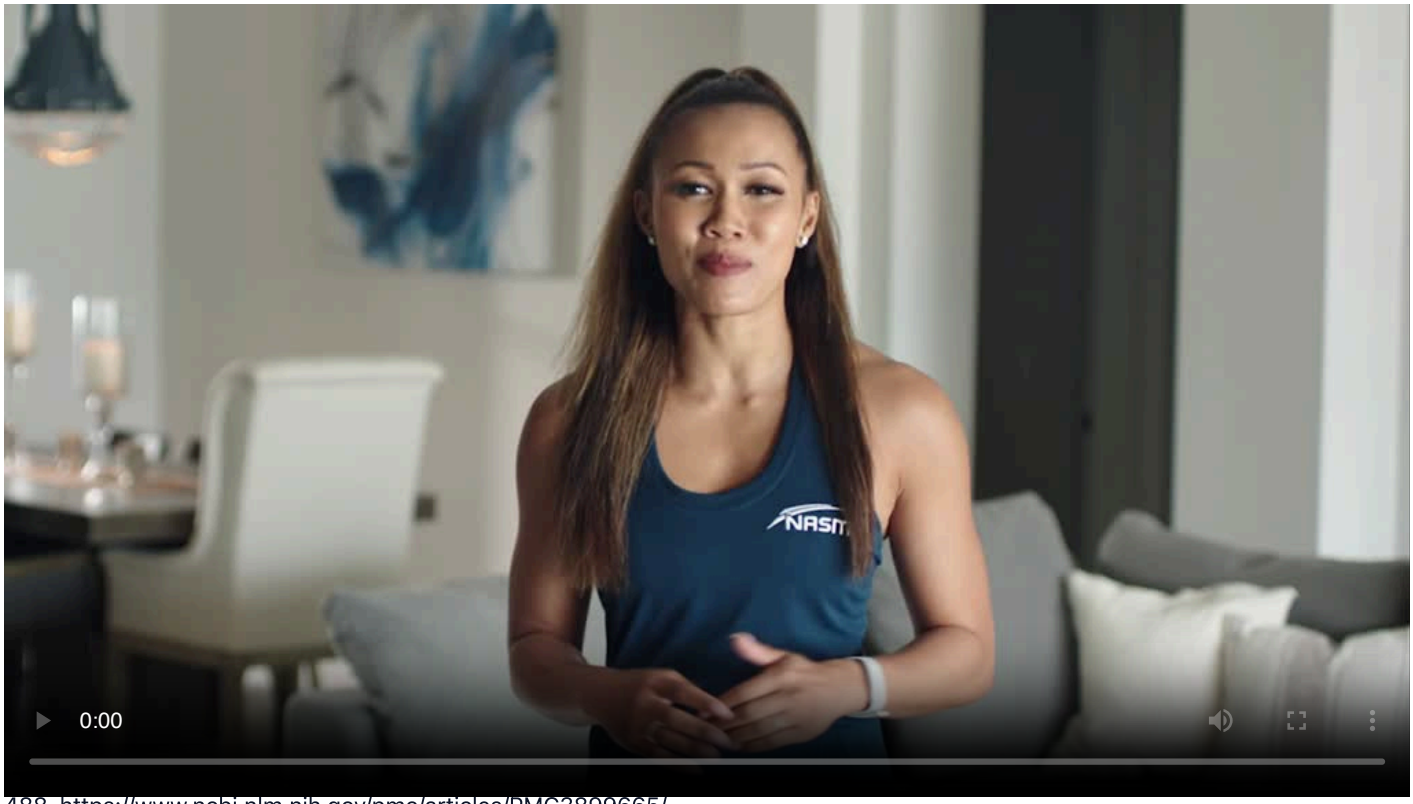
Lastly, clients want to know that they are being supported and monitored by a fitness professional they can trust. By integrating wearable technology into virtual coaching, fitness professionals have the ability to further individualize recommendations and continuity of care with 24/7 monitoring beyond what can be offered in a training facility.

Positive reinforcement

A stimulus following a behavior that increases the likelihood of an individual continuing that behavior – For example, improvements in scale weight due to nutritional intervention increase the likelihood of an individual continuing to adhere to the nutrition program.

# Summary

Biometric data provided by wearable technologies provides fitness professionals with reliable data to assist in developing, modifying, and continuously monitoring virtual programs for client compliance and outcomes. Many of these biometrics, including sleep and heart rate variability, may be applied to support a better-quality program for all client types based on their relationship with physical health, mental health, risk of injury, and athletic performance. When integrated properly, wearable technology provides the fitness professional with the ability to facilitate a 24/7 monitoring approach to improve client outcomes while also optimizing their ability to scale their virtual coaching business. Please refer to the video below for a summary of the topics discussed within this chapter.




NASM VCS Lecture CH7
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488. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3899665/

Centers for Disease Control. (2020). Sleep and Sleep Disorders. Retrieved from https://www.cdc.gov/sleep/index.html

Cleveland Clinic. (2019). Vital Signs. Retrieved from https://my.clevelandclinic.org/health/articles/10881-vital-signs

Colton, H. R., & Altevogt, B. M. (Eds.). (2006). *Sleep disorders and sleep deprivation: An unmet public health problem*. Washington, DC: National Academies Press.

Cretikos, M. A., Bellomo, R., Hillman, K., Chen, J., Finfer, S., & Flabouris, A. (2008). Respiratory rate: the neglected vital sign. *Medical Journal of Australia*, *188*(11): 657–659. https://doi.org/10.5694/j.1326-5377.2008.tb01825.x

Dattilo, M., Antunes, H. K. M., Medeiros, A., Mônico Neto, M., Souza, H. S., Tufik, S., & de Mello, M. T. (2011). Sleep and muscle recovery: Endocrinological and molecular basis for a new and promising hypothesis. *Medical Hypotheses*, *77*(2): 220–222. https://doi.org/10.1016/j.mehy.2011.04.017

Dong, J.-G. (2016). The role of heart rate variability in sports physiology. *Experimental and Therapeutic Medicine*, *11*(5): 1531–1536. https://doi.org/10.3892/etm.2016.3104

Expert Program Management. (2018). Reinforcement Theory of Motivation. Retrieved from https://expertprogrammanagement.com/2018/10/reinforcement-theory-of-motivation/

Fang, S.-C., Wu, Y.-L., & Tsai, P.-S. (2019). Heart Rate Variability and Risk of All-Cause Death and Cardiovascular Events in Patients With Cardiovascular Disease: A Meta-Analysis of Cohort Studies. *Biological Research for Nursing*, *22*(1): 45–56. https://doi.org/10.1177/1099800419877442

Hafen, B. B., & Sharma, S. (2020). *Oxygen saturation*. Treasure Island, Fl: StatPearls.

Holland, K. (2019). Is My Blood Oxygen Level Normal? Retrieved from https://www.healthline.com/health/normal-blood-oxygen-level#:%7E:text=Normal%3A%20A%20normal%20ABG%20oxygen,these%20ranges%20may%20not%20apply

Izumi, I., Nasermoaddeli, A., Sekine, M., & Kagamimori, S. (2005.). Effect of moderate alcohol intake on nocturnal sleep respiratory parameters in healthy middle-aged men. *Environmental Health and Preventive Medicine*, *10*(1): 16–20. https://doi.org/10.1265/ehpm.10.16

Jandackova, V. K., Scholes, S., Britton, A., & Steptoe, A. (2016). Are Changes in Heart Rate Variability in Middle-Aged and Older People Normative or Caused by Pathological Conditions? Findings From a Large Population-Based Longitudinal Cohort Study. *Journal of the American Heart Association*, *5*(2): e002365. https://doi.org/10.1161/jaha.115.002365

Javaloyes, A., Sarabia, J. M., Lamberts, R. P., & Moya-Ramon, M. (2019). Training Prescription Guided by Heart-Rate Variability in Cycling. *International Journal of Sports Physiology and Performance*, *14*(1): 23–32. https://doi.org/10.1123/ijspp.2018-0122

Liu, Y.-Z., Wang, Y.-X., & Jiang, C.-L. (2017). Inflammation: The Common Pathway of Stress-Related Diseases. *Frontiers in Human Neuroscience*, *11*: 316. https://doi.org/10.3389/fnhum.2017.00316

Maeda, Y., Sekine, M., & Tamura, T. (2011). The Advantages of Wearable Green Reflected Photoplethysmography. *Journal of Medical Systems*, *35*(5): 829–834. https://doi.org/10.1007/s10916-010-9506-z

Nall, R. (2017). A Mom's Guide to Pediatric Vital Signs. Retrieved from https://www.healthline.com/health/pediatric-vital-signs

National Institute of General Medical Sciences. (2020). Circadian Rhythms. Retrieved from https://www.nigms.nih.gov/education/fact-sheets/Pages/circadian-rhythms.aspx#:%7E:text=Circadian%20rhythms%20are%20physical%2C%20mental,a%20light%2Drelated%20circadian%20rhythm

National Institute of Neurological Disorders and Stroke. (2019). Brain Basics: Understanding Sleep. Retrieved from https://www.ninds.nih.gov/disorders/patient-caregiver-education/understanding-sleep

National Sleep Foundation. (2015). National Sleep Foundation Recommends New Sleep Times. Retrieved from https://www.sleepfoundation.org/press-release/national-sleep-foundation-recommends-new-sleep-times

Nauman, J., Aspenes, S. T., Nilsen, T. I. L., Vatten, L. J., & Wisløff, U. (2012). A Prospective Population Study of Resting Heart Rate and Peak Oxygen Uptake (the HUNT Study, Norway). *PLoS ONE*, *7*(9): e45021. https://doi.org/10.1371/journal.pone.0045021

Ozdal, M., Pancar, Z., Çinar, V., & Bilgiç, M. (2017). Effect of Smoking on Oxygen Saturation in Healthy Sedentary Men and Women. *EC Pulmonology and Respiratory Medicine*, *4*(6): 178–182.

Pichot, V., Roche, F., Gaspoz, J. M., Enjolras, F., Antoniadis, A., Minini, P., Barthélémy, J. C. (2000). Relation between heart rate variability and training load in middle-distance runners. *Medicine & Science in Sports & Exercise*, *32*(10): 1729–1736. https://doi.org/10.1097/00005768-200010000-00011

Polar. (2020). Running heart rate zones: The basics. Retrieved from https://www.polar.com/blog/running-heart-rate-zones-basics/

Reimers, A. K., Knapp, G., & Reimers, C.-D. (2018). Effects of Exercise on the Resting Heart Rate: A Systematic Review and Meta-Analysis of Interventional Studies. *Journal of Clinical Medicine*, *7*(12): 503. https://doi.org/10.3390/jcm7120503

Scheiermann, C., Kunisaki, Y., & Frenette, P. S. (2013). Circadian control of the immune system. *Nature Reviews Immunology*, *13*(3): 190–198. https://doi.org/10.1038/nri3386

Seccareccia, F., Pannozzo, F., Dima, F., Minoprio, A., Menditto, A., Lo Noce, C., & Giampaoli, S. (2001). Heart Rate as a Predictor of Mortality: The MATISS Project. *American Journal of Public Health*, *91*(8): 1258–1263. https://doi.org/10.2105/ajph.91.8.1258

Shaffer, F., McCraty, R., & Zerr, C. L. (2014). A healthy heart is not a metronome: an integrative review of the heart's anatomy and heart rate variability. *Frontiers in Psychology*, *5*: 1040. https://doi.org/10.3389/fpsyg.2014.01040

Shcherbina, A., Mattsson, C. M., Waggott, D., Salisbury, H., Christle, J. W., Hastie, T., Wheeler, M. T., & Ashley, E. A. (2017). Accuracy in Wrist-Worn, Sensor-Based Measurements of Heart Rate and Energy Expenditure in a Diverse Cohort. *Journal of Personalized Medicine*, *7*(2): 3. https://doi.org/10.3390/jpm7020003

Spigulis, J., Gailite, L., Lihachev, A., & Erts, R. (2007). Simultaneous recording of skin blood pulsations at different vascular depths by multiwavelength photoplethysmography. *Applied Optics*, *46*(10): 1754–1759. https://doi.org/10.1364/ao.46.001754

Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology (1996). Heart Rate Variability. *Circulation*, *93*(5): 1043–1065. https://doi.org/10.1161/01.cir.93.5.1043

Vogels, E. A. (2020. About one-in-five Americans use a smart watch or fitness tracker. Retrieved from https://www.pewresearch.org/fact-tank/2020/01/09/about-one-in-five-americans-use-a-smart-watch-or-fitness-tracker/

Williams, S., Booton, T., Watson, M., Rowland, D., & Altini, M. (2017). Heart Rate Variability is a Moderating Factor in the Workload-Injury Relationship of Competitive CrossFit Athletes. *Journal of Sports Science & Medicine*, *16*(4): 443–448. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5721172/

Xie, J., Wen, D., Liang, L., Jia, Y., Gao, L., & Lei, J. (2018). Evaluating the Validity of Current Mainstream Wearable Devices in Fitness Tracking Under Various Physical Activities: Comparative Study. *JMIR mHealth and uHealth*, *6*(4): e94. https://doi.org/10.2196/mhealth.9754

Young, H. A., & Benton, D. (2018). Heart-rate variability: a biomarker to study the influence of nutrition on physiological and psychological health? *Behavioural Pharmacology*, *29*:9(2): 140–151. https://doi.org/10.1097/fbp.0000000000000383

Zafar, H., Albarrati, A., Alghadir, A. H., & Iqbal, Z. A. (2018). Effect of Different Head-Neck Postures on the Respiratory Function in Healthy Males. *BioMed Research International*, *2018*(5): 1–4. https://doi.org/10.1155/2018/4518269